Actividad Integradora-Taller de Redes

CURSO : 4to 9na
ALUMNOS : Alba Lautaro y Gaitán Bautista

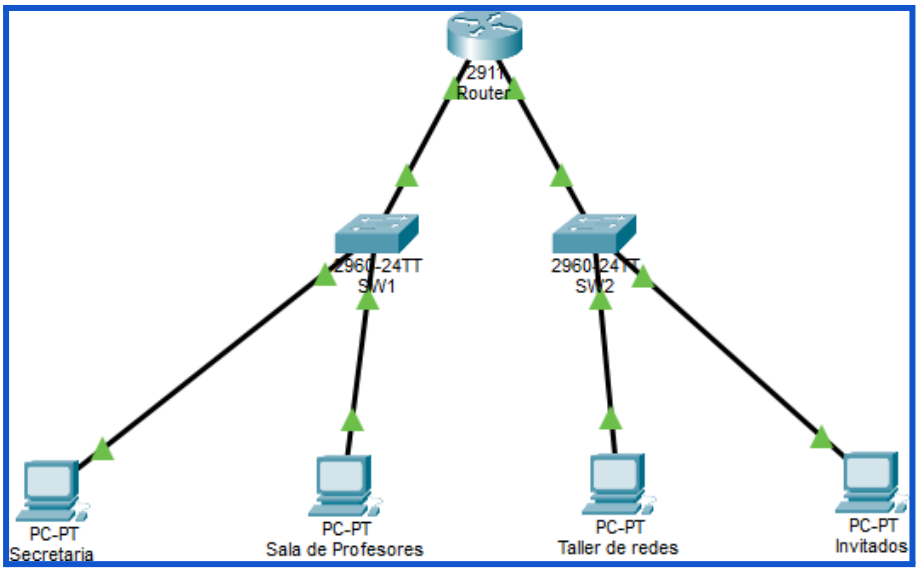


Tabla de Direccionamientos :

PC	Puerto	IPv4	VLAN
Secretaria	SW1 - Fa0/1	192.168.10.10/24	10
Sala de Profesores	SW1 - Fa0/2	192.168.20.10/24	20
Taller de Redes	SW2 - Fa0/1	192.168.30.10/24	30
Sala de Invitados	SW2 - Fa0/2	192.168.40.10/24	40
Router		**IPv4**	**Máscara**
Router 2911	SW1 - G0/0.10	192.168.10.1	255.255.255.0
Router 2911	SW1 - G0/0.20	192.168.20.1	255.255.255.0
Router 2911	SW2 - G0/0.30	192.168.30.1	255.255.255.0
Router 2911	SW2 - G0/0.40	192.168.40.1	255.255.255.0

Configuraciones Claves:

Switch1 - Configuración de VLA

enable
configure terminal

vlan 10
name Secretaria
exit

vlan 20
name Profesores
exit

Configuración de PC de secretaria

interface range fa0/1 switchport
mode access switchport access
vlan 10 exit

Configuración de PC de sala de profesores

interface range fa0/2 switchport
mode access switchport access
vlan 20 exit

Configuración trunk para Router

interface gigabitEthernet0/1
switchport mode trunk
exit

Switch2
Configuración de VLANs

vlan 30
name Taller
exit

vlan 40
name Invitados
exit

Configuración de PC de Taller de redes

interface range fa0/1 switchport
mode access switchport access
vlan 30 exit

Configuración de PC para sala de invitados

interface range fa0/2 switchport
mode access switchport access
vlan 40 exit

Configuración trunk para Router

interface gigabitEthernet0/1
switchport mode trunk
exit

Router
Configuración de VLANs 10, 20, 30 y 40

enable
configure terminal

interface gigabitEthernet0/0 no
shutdown
exit

interface gigabitEthernet0/0.10 encapsulation dot1Q
10
ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet0/0.20 encapsulation dot1Q
20
ip address 192.168.20.1 255.255.255.0
exit

interface gigabitEthernet0/2.30 encapsulation dot1Q
30
ip address 192.168.30.1 255.255.255.0
exit

interface gigabitEthernet0/2.40 encapsulation dot1Q
40
ip address 192.168.40.1 255.255.255.0
exit

Conectividad entre sectores :

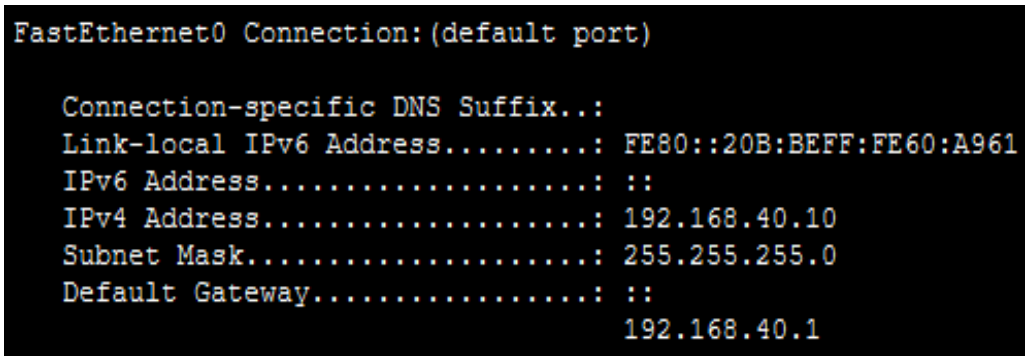
	Secretaria
ping	C:\>ping 192.168.10.10 Pinging 192.168.10.10 with 32 bytes of data: Reply from 192.168.10.10: bytes=32 time<1ms TTL=127 Reply from 192.168.10.10: bytes=32 time<1ms TTL=127 Reply from 192.168.10.10: bytes=32 time<1ms TTL=127 Reply from 192.168.10.10: bytes=32 time<1ms TTL=127 Ping statistics for 192.168.10.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
tracert	C:\>tracert 192.168.10.10 Tracing route to 192.168.10.10 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 192.168.40.1 2 0 ms 0 ms 0 ms 192.168.10.10 Trace complete.
ipconfig	C:\>ipconfig FastEthernet0 Connection:(default port) Connection-specific DNS Suffix..: Link-local IPv6 Address.....: FE80::20A:41FF:FE5C:4E8 IPv6 Address.....: :: IPv4 Address.....: 192.168.10.10 Subnet Mask.....: 255.255.255.0 Default Gateway.....: :: 192.168.10.1

	Sala de profesores
ping	C:\>ping 192.168.20.10 Pinging 192.168.20.10 with 32 bytes of data: Reply from 192.168.20.10: bytes=32 time<1ms TTL=127 Reply from 192.168.20.10: bytes=32 time<1ms TTL=127 Reply from 192.168.20.10: bytes=32 time<1ms TTL=127 Reply from 192.168.20.10: bytes=32 time<1ms TTL=127 Ping statistics for 192.168.20.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
tracert	C:\>tracert 192.168.20.10 Tracing route to 192.168.20.10 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 192.168.10.1 2 1 ms 0 ms 0 ms 192.168.20.10 Trace complete.
ipconfig	C:\>ipconfig FastEthernet0 Connection:(default port) Connection-specific DNS Suffix..: Link-local IPv6 Address.....: FE80::201:42FF:FE75:2EA5 IPv6 Address.....: :: IPv4 Address.....: 192.168.20.10 Subnet Mask.....: 255.255.255.0 Default Gateway.....: :: 192.168.20.1

	Taller de redes
ping	C:\>ping 192.168.30.10 Pinging 192.168.30.10 with 32 bytes of data: Reply from 192.168.30.10: bytes=32 time<1ms TTL=127 Reply from 192.168.30.10: bytes=32 time<1ms TTL=127 Reply from 192.168.30.10: bytes=32 time<1ms TTL=127 Reply from 192.168.30.10: bytes=32 time<1ms TTL=127 Ping statistics for 192.168.30.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms

tracert	C:\>tracert 192.168.30.10 Tracing route to 192.168.30.10 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 192.168.10.1 2 1 ms 0 ms 0 ms 192.168.30.10 Trace complete.
ipconfig	C:\>ipconfig FastEthernet0 Connection:(default port) Connection-specific DNS Suffix..: Link-local IPv6 Address.....: FE80::201:43FF:FEE1:44C1 IPv6 Address.....: :: IPv4 Address.....: 192.168.30.10 Subnet Mask.....: 255.255.255.0 Default Gateway.....: :: 192.168.30.1

	Sala de Invitados
ping	C:\>ping 192.168.40.10 Pinging 192.168.40.10 with 32 bytes of data: Reply from 192.168.40.10: bytes=32 time<1ms TTL=127 Reply from 192.168.40.10: bytes=32 time<1ms TTL=127 Reply from 192.168.40.10: bytes=32 time<1ms TTL=127 Reply from 192.168.40.10: bytes=32 time<1ms TTL=127 Ping statistics for 192.168.40.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
tracert	C:\>tracert 192.168.40.10 Tracing route to 192.168.40.10 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 192.168.10.1 2 0 ms 0 ms 0 ms 192.168.40.10 Trace complete.

ipconfig	

```
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::20B:BEFF:FE60:A961
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.40.10
   Subnet Mask.....: 255.255.255.0
   Default Gateway.....: ::
                                 192.168.40.1
```

PUNTOS QUE NO PUDIMOS REALIZAR :

5. Justificar y aplicar un criterio de calidad de servicio (QoS): priorizar, por ejemplo, el tráfico del Taller de Redes por sobre el de la Sala de Invitados.

6. Identificar qué dispositivos de la topología requerirían alimentación Power over Ethernet (por ejemplo, el access point) y fundamentar la elección.

7. Configurar el punto de acceso inalámbrico: SSID, cifrado WPA2 o WPA3, contraseña robusta y aislamiento de clientes para la red de invitados.

ESTO POR FALTA DE CONOCIMIENTO
